

Trainings ▾

General Information

The C Command Elite™ and C Command Elite Plus™ is the premium version of the customer interface box (CIB).

- The CIB is a common point for engine and vessel connections.
- The CIB provides connections to the ECM through Deutsch™ connectors at the bottom of the CIB panel.
- Communication protocols are opened through SAE J1939 Ethernet and Modbus™ connections. This communication allows for accurate updates of engine statistics.
- No engine room panel (ERP) is required because the CIB functions as a local control panel.
- Supports all C Command™, C Command Elite™, and C Command Elite Plus™ remote options.
- Full color displays on the diesel control unit (DCU410) and remote panel (RP).
- Remote panel touch screens capable of monitoring up to 8 engines.
- Capable of integrating 6 vessel supplied inputs.
- The C Command Elite™ and C Command Elite Plus™ panel systems support feature meets Marine Society requirements for Class approvals.
- The DCU410 includes modules for safety and alarm system requirements.
- Includes sensors, switches, and harnesses to meet Type Approved requirements. Additional vessel inputs available to meet requirements are **not** yet supported.

The combination of these features provides a very flexible and stable platform for engine control and diagnostics.

Customer Interface Box

Isolators are required for the installation of the CIB panel. Engine mounted panels are **not** required.

The CIB contains the most basic functions which are required by all system configurations. Most functions in the CIB are performed by the CIB integrated DCU410 control panel and the CIB logic unit (CLU). The DCU410 is a microprocessor controlled unit that performs the extended functions of the C Command Elite™ and C Command Elite Plus™ panel systems. The CLU contains relays, diodes, resistors, switches, microcontroller, and power management capabilities.

The CIB is a mandatory option that is delivered with every engine, and is designed to function when connected to the vessel, whether or **not** a Cummins Inc. provided remote panel is supplied.

The CIB is capable of accepting power from two independent power supplies, but **only** one is required for normal operation. The second power supply is intended to be a backup for the primary power supply in the event of a primary power supply malfunction or to fulfill marine

society requirements in the case of C Command Elite Plus™. The CLU automatically switches between the primary and secondary power supplies with any primary malfunction. In addition to the power supply switching, the CIB also conditions and distributes the power received by the vessel power supply. It is capable of detecting under-voltage, over-voltage, under current, over-current, and reverse current for each of the vessel power inputs.

⚠ CAUTION ⚠

The CIB is not protected from AC power sources. Damage can occur if AC power is applied to the CIB.

The primary and secondary power supplies are also isolated from each other in the circuitry of the CIB. This makes sure there is protection from ground faults and galvanic issues. The CIB power management system is designed to supply power for **only** Cummins Inc. supplied devices. Vessel connections are primary connected to the X4 terminal strip inside the CIB.

The C Command Elite Plus™ panel system requires both power supplies to be connected.

The Ethernet switch is included in the CIB for remote panel and vessel networking connections.

Cummins Inc. supplies the sensors and connections for the C Command Elite Plus™ panel system.

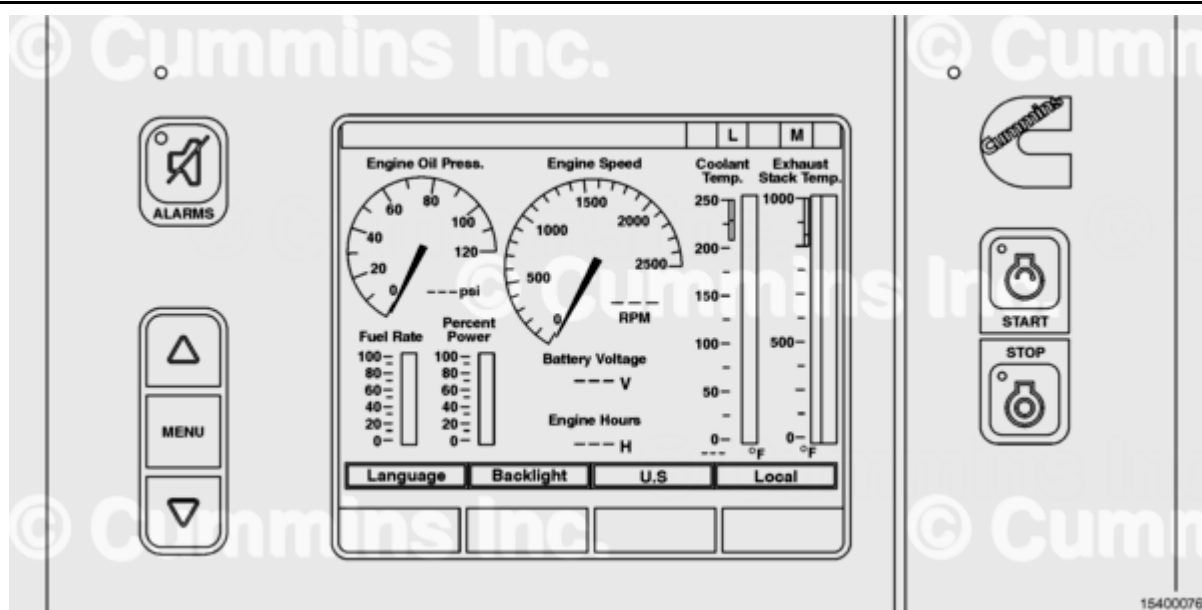
The CIB is connected to the engines ECM through Deutsch™ connectors C8, C9, C10, and data link.

Note : Power for customer supplied devices is not allowed to be connected to the CIB vessel terminal strip.

The connections are as follows:

- ECM unswitched power
- ECM switched power
- J1939 data link
- Diagnostic lamp outputs
- Analog outputs (speed, pressure, and temperature)
- Speed control
- Idle control
- Remote sensors.

Diesel Control Unit



CIB Integrated Control Panel Diesel Control Unit (DCU)

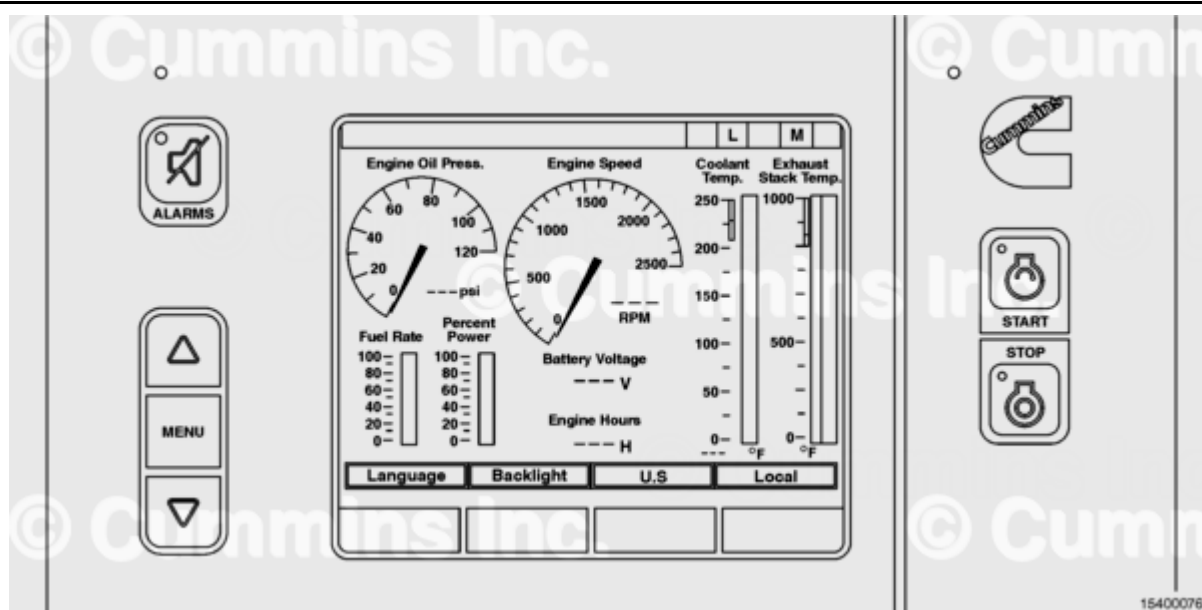
All control and monitoring can be achieved at the DCU full-color display on the CIB. No separate Engine Room Panel is necessary or provided. See the illustration for a close-up of the DCU unit.

The DCU communicates with the ECM via the J1939 bus to the ECM. This display allows the operator to view a selection of parameters as well as full text descriptions of ECM fault codes. The C Command™ ED-3 (if equipped) **must** be configured as described for C Command™ parameters if connected to the C Command Elite™ and C Command Elite Plus™ system. Engine command and monitoring is controlled via both hard and soft keys located on the DCU unit display.

The DCU outputs all signals to the CIB Ethernet switch, the Ethernet output is then routed to any Remote Panels that are present on the vessel. Each DCU will be assigned an Engine Number (IP) for recognition on the vessel network. The configuration of this number will be requested the first time the DCU is powered up. This number can also be changed from the DCU main menu, if necessary. In addition, with the use of the web-based configuration interface, the DCU can also be assigned a text name.

Any additional inputs from the vessel need to be configured at the DCU unit. The DCU unit controls what is displayed on the remote panel screens.

Shutdown Unit



CIB Integrated Control Panel Remote Input/Output Unit (RIO)

The SDU unit is an independent component within the CIB.

There are LED indicators on the unit face for status and fault codes.

Switch conditions are available on remote panels as well as on Modbus™ and data link. The Reset button on the unit face acknowledges faults and enables shutdown functionality.

The Safety System requirement from the marine agencies is met with the use of the Shutdown Unit (SDU). The SDU provides secondary engine protection by shutting down the engine if any critical parameters are exceeded. With the exception of engine speed, these signals are provided to the SDU as switch closures based on the exceeded condition.

The SDU has five (5) Cummins® dedicated channels for Safety System shutdown use. An additional three (3) other switch inputs are provided for vessel initiated shutdowns.

The SDU communicates system faults through the use of amber LEDs for individual input channels. Red LEDs are used to communicate a parameter has been exceeded and an in-process engine shutdown. A Shutdown Override function is available on the SDU. This function disables all engine protections provided by the SDU, with the exception of Engine Overspeed protection. The SDU may be configured so that other protections will continue to be enabled when the Shutdown Override function is in use.

A Reset function is included as part of the SDU as well. In the event of a SDU based shutdown, the SDU **must** be reset in order to allow for engine re-start. This function can be utilized by the Reset button located on the SDU front face.

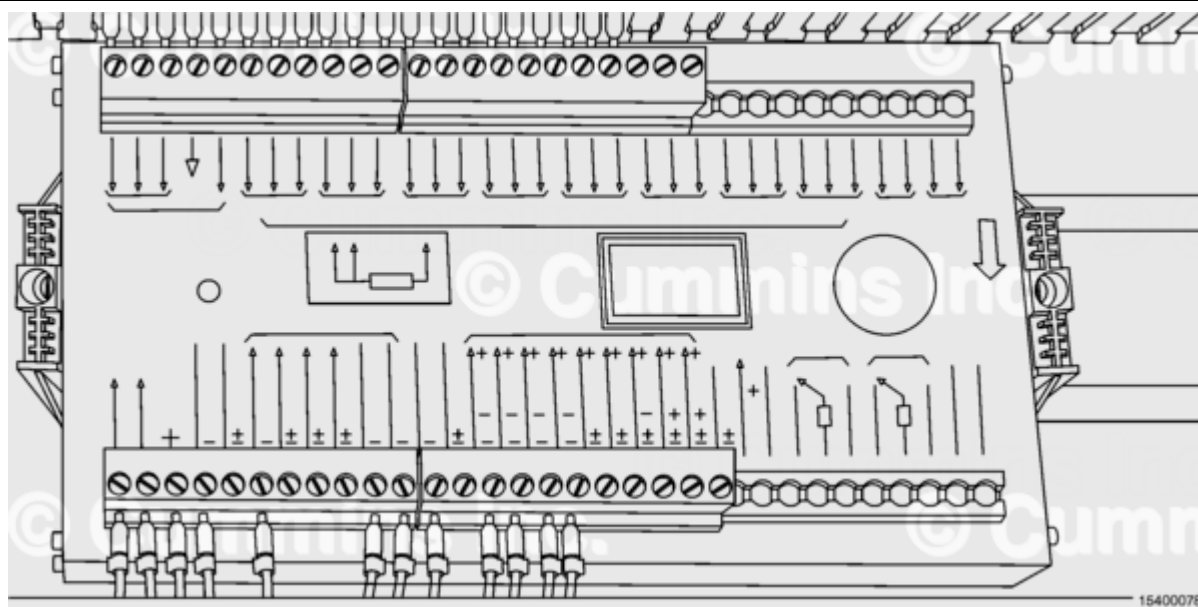
All QSK Series engines with MCRS are equipped with the same sensors/switches for the Safety System. The following table is a description of the SDU channels and their function.

SDU Channel Descriptions		
Quantity	Type	Description
2	Speed Sensor	Overspeed Speed Range for Switches

SDU Channel Descriptions

Quantity	Type	Description
5	Switch Input (normally open)	Cummins Dedicated Shutdowns: Coolant Temperature, Coolant Pressure, Lubricating Oil Pressure (low-speed range), Lubricating oil Pressure (high-speed range), Remote Engine Stop
3	Switch (normally open)	Additional Vessel Shutdowns
2	Switch (normally open)	Shutdown Override Contact Reset Button
Note: Three (3) shutdown input channels are available.		

Remote Input/Output Unit



The Remote Input Output Unit (RIO) is an analog-to-digital converter for alarm sensors. This unit is contained within the CIB and allows for the integration of alarm sensor inputs into the panel system. The RIO broadcasts analog input signals from on-engine alarm sensors and any additional vessel alarms sensor via Modbus™, which can then be read by the DCU.

Note : The information from the sensors connected to this unit is **only** available on the Ethernet, Modbus™, or CANOpen interfaces. They can **not** be viewed with a Calterm or INSITE™ electronic service tool.

The required Alarm and Monitoring system sensors to meet Marine Society classifications from Cummins are wired directly from the CIB connector C4 to the RIO. Additional RIO channels are made available at the RIO for vessel supplied sensors that may be required to meet Marine

Society requirements **not** covered by the C Command Elite Plus™ system. A list of the Cummins® dedicated channels, per engine type, and the channels available for vessel use are outlined in the Cummins® Dedicated Alarm and Monitoring System Channels table.

The connections required for the RIO additional channels for the vessel **must** be made at the RIO directly. The terminal numbers for each vessel channel can be found on the C Command Elite Plus™ Marine Wiring Diagram. These channels can be configured from either the DCU Configuration Menu or the web-browser based interface. From there, channels can be activated, sensor ranges/switch conditions can be set, and alarm thresholds can be defined.

Cummins® Dedicated Alarm and Monitoring System Channels

	Engine Source		
Description	QSK19	QSK38	QSK50/QSK60
Fuel Leakage (normally closed)	R	R	R
Intake Manifold Temperature (left back front on KV engine)	E	E	E
Left Bank Rear Intake Manifold Temperature (KV engines only)		R	E
Right Bank Front Intake Manifold Temperature (KV engines only)		R	E
Right Bank Rear Intake Manifold Temperature (KV engines only)		R	E
Spare Temperature (e.g., Gear Oil Temperature)	R	R	R
Coolant Pressure	R	R	E
Lubricating Oil Filter Inlet Pressure	R	R	E
Sea Water Pressure	R	R	R
Spare Pressure (e.g., Air Starter)	R	R	R

LEGEND: R = RIO Monitored, E = ECM Monitored

Additional Alarm and Monitoring System Channels for Vessel Supplied Sensors			
	Engine Source		
Description	QSK19	QSK38	QSK50/QSK60
Three (3) Temperature Channels	R	R	R
Five (5) Pressure Channels	R	R	R
One (1) Coolant Level Channel	R	R	R
Two (2) Exhaust Temperature Channels	R	R	R
Three (3) Switch Inputs (normally open)	R	R	R
Two (2) Relay Outputs	R	R	R
Two (2) Switch Outputs	R	R	R
LEGEND: R = RIO Monitored, E = ECM Monitored			
RIO Channel Descriptions			
Quantity	Type	Description	
4	Switch Input	24-VDC Configurable NO/NC Switch Input	
8	PT100	Temperature Sensor Input	
9	4-20 mA	Pressure Sensor Input	
1	0-5 VDC	Coolant Level Sensor Input (Ratiometric)	
2	Type K	Thermocouple Input	
2	Relay Output	Configurable Relay Switch Output	
2	Switch Output	24-VDC Configurable NO/NC Switch Output	
Remote Panel			

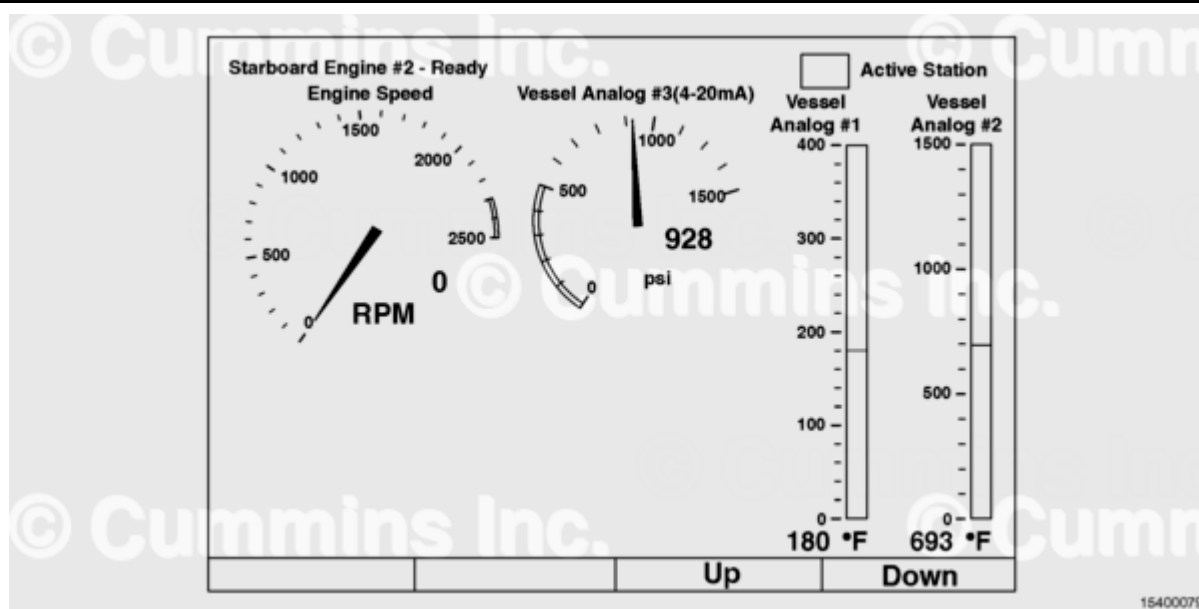
The remote panel **must** be mounted in a protected enclosure or console. The remote panel installation kit includes all hardware and seals.

The remote panel (RP) is the main engine display available on the C Command Elite™ and C Command Elite Plus™ panel systems. Alarms are displayed as on the DCU410 unit and include a full text description of the alarm or fault. The remote panel includes all of the functionality available at the DCU410 unit, including engine start and stop.

Remote Panel Features:

- Automatically scans for connected engines (DCUs)
- Up to 8 DCUs may be connected to one RP via Ethernet network
- Displays engine parameters as displayed on each DCU (Content dependent on DCU)
- Capability to Start and Stop multiple engines from a single RP
- Capability to display Alarm status for multiple engines from a single RP
- Displays service interval timer, as on DCU
- Alarm broadcasting to the RP may be disabled (in the case of a manned engine room).

Like the DCU, the functions of the RP are accessed with a combination of Hard and Soft Keys. The difference is that the RP has a touch screen interface, so rather than pressing buttons below a Soft option, the operator **must** press the function text directly. The Hard keys function in the same manner as the DCU, with **only** one difference. A Home Hard key has been added to ease navigation through the touch screen pages. Also like the DCU, RP **must** be assigned an IP number in order to properly be recognized by the vessel network. This unique IP number will allow for several RPs to be installed on a vessel network without conflict. Additionally, each RP can be assigned a station name to more easily distinguish the location of each RP and which station is in present control. The remote panel can be used to view the status of any of the engines that it has been set up to monitor, without any interaction with other remote panels in the vessel network. To perform control functions, the remote panel **must** be in Active Station Mode.

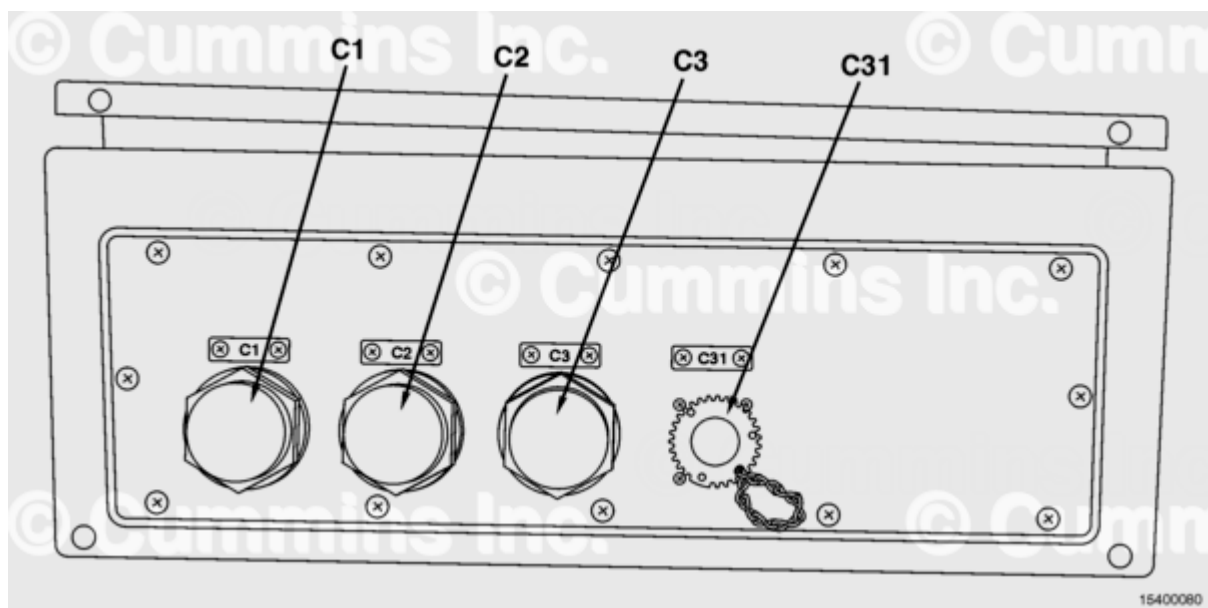


The remote panel can be assigned a panel location. The panel location is selected from a list in the remote panel system configuration menu. The location will be shown in the upper right hand corner of the display to denote which location is the active station. The following are remote panel locations:

- Bridge
- Port wing station
- Starboard wing station
- Tower
- Engine control room
- Aft station
- Bow station
- Other.

New Connection

Diesel Control Unit



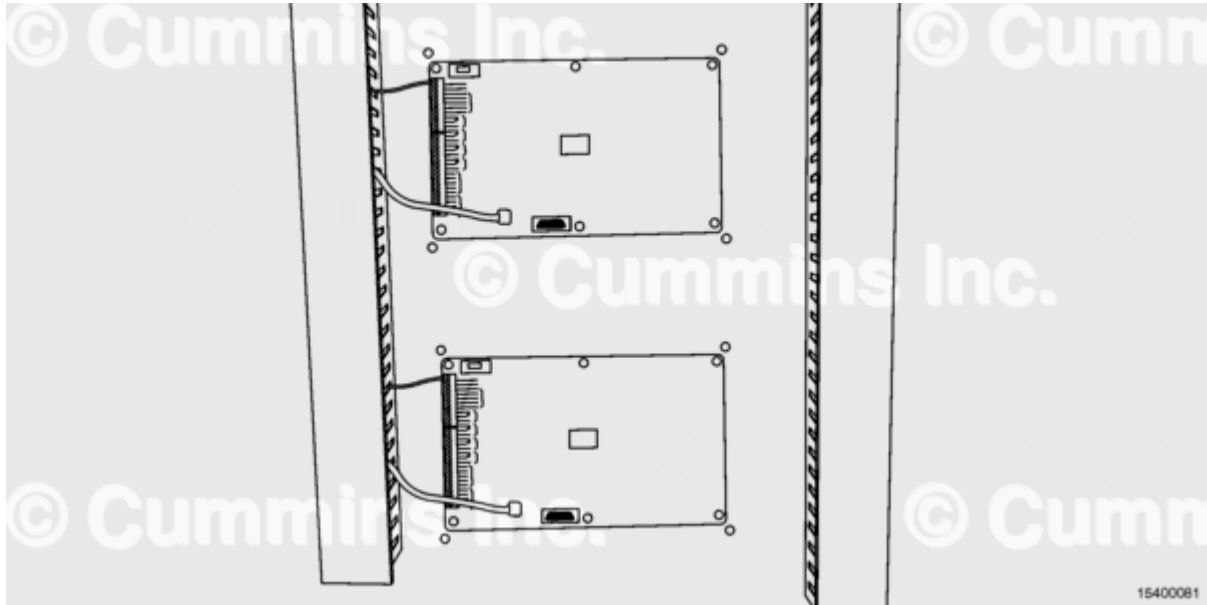
Connectors at Bottom of CIB

The C Command™ CIB has the following connectors at the bottom of the cabinet:

- "C1" - 16-Pin Deutsch™ Connector
- "C2" - 23-Pin Deutsch™ Connector
- "C3" - 31-Pin Deutsch™ Connector
- "C31" - Service Connector

Connector C1 is connected to Connector C8 (16-pin) on the engine. Connector C2 is connected to Connector C9 (23-pin) on the engine. Connector C3 on the CIB is connected to C10 (31-pin) on the Engine.

Remote Panel



Connection to the remote panel **must** be made through the Ethernet connection so the remote can communicate with the DCU.

Setup

Diesel Control Unit

The following is an overview of the DCU Power Wizard process.

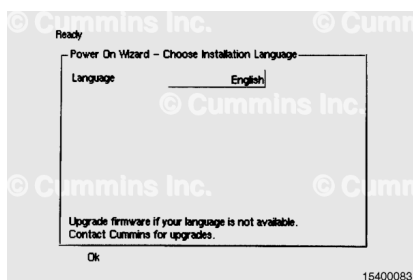
The DCU will power up in conjunction with the CIB Power Switch. During the first power up, the DCU will enter the Power On Wizard. The panel system operates on 24-VDC.

The setup wizard will prompt the user to input the basic data necessary to configure the DCU for the engine type, engine location, and application. The wizard will also allow the operator to set a custom password to protect higher level settings to be changed.



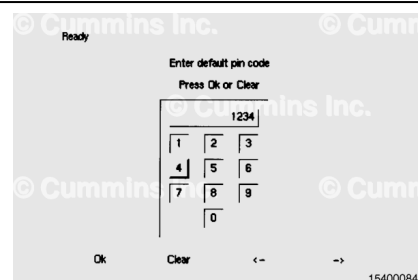
1. Select a language:

- a. Choose from available languages.
- b. Press OK to confirm.



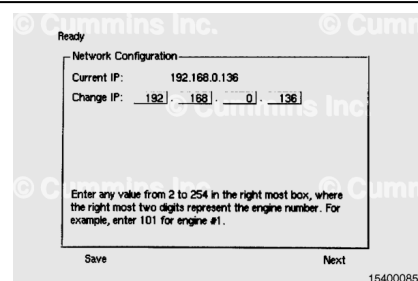
2. Choose administrator password:

- Input default password (1234).
- Input new password.
- Re-enter new password for validation.
- Press OK to confirm.



3. Select an IP number:

- View default IP number for the DCU.
- Change the value to required IP for vessel network.



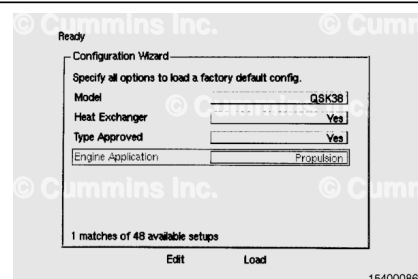
Note : Devices can not have identical IP addresses in an Ethernet network.

- Press Save to commit the value.
- Press Exit to go to the next step.

This value becomes the engine identification number for the remote panel. A name can also be assigned to the engine with the web-browser based tool.

4. Load Configuration:

- Highlight "Model" selection box.
- Press Edit to change the value.
- Use Arrow soft keys to move between options.
- Press OK to save change.
- Press Down Arrow hard key to highlight the next selection.
- Repeat steps a through d for "Heat Exchanger", "Type Approved", and "Engine Application".
- Press the Load soft key to commit the configuration to the DCU.
- Press Yes to load the new configuration or press Cancel to return to the setup menu.



After configuration, the operator should return to the Home page on the DCU unit.

Remote Panel

The following is an overview of the Remote Panel (RP) Power Wizard process.

The remote panel will also power up in conjunction with the CIB Power Switch.

During the first power up, the RP will also enter its Power On Wizard.

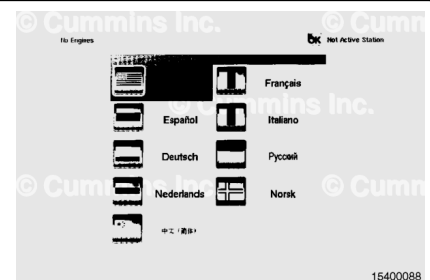
The setup wizard will prompt the user to input the basic data necessary to configure the RP for the Language, Location, IP Address, and Monitored Engine(s). It will also allow the user to set a custom password to protect higher level settings to be changed.

The RP Power On Wizard is very similar to the DCU Power On Wizard. The main difference is that most of the options for the RP are set by DCU configuration. The information that is required for initial setup is much less than the DCU.



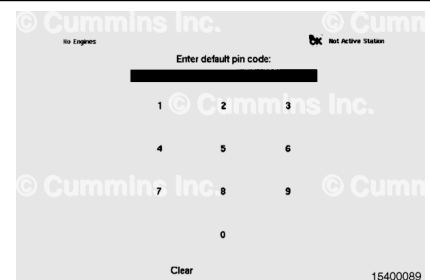
1. Select a language:

- Choose from available languages.
- Press OK to confirm.



2. Choose administrator password:

- Input default password (1234).
- Input new password.
- Re-enter new password for validation.
- Press OK to confirm.



3. Select an IP number:

- View default IP number for the DCU.
- Change the value to required IP for vessel network.

Note : Devices can not have identical IP addresses in an Ethernet network.

c. Press Save to commit the value.

d. Press Exit to go to the next step.

This value becomes the engine identification number for the remote panel. A name can also be assigned to the engine with the web-browser based tool.

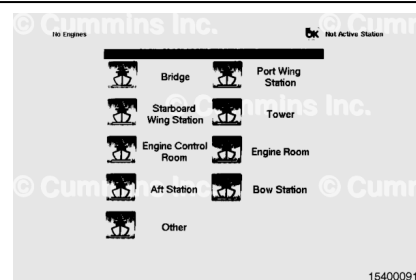


4. Select panel location name:

a. Choose from available panel names.

b. Press OK to confirm.

After configuration, the operator should return to the Home page on the DCU unit.



5. Select engine(s) to monitor:

a. Available engine(s) will be listed by IP number and engine name, if set.

b. Check select box for engine(s) that RP is intended to monitor.

c. Press OK to confirm.

Note : Presently only single engine monitoring is available. Multiple engine monitoring is planned.



6. Remote panel complete:

a. Cummins Inc. RP information screen will be shown on the display.

b. RP is ready for system commands.

After configuration, the operator should return to the Home page on the DCU unit.



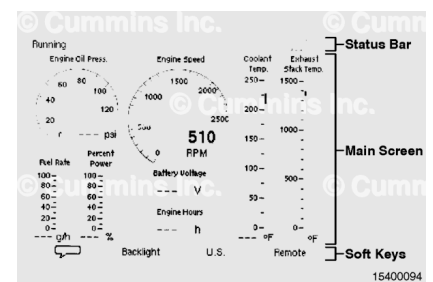
Viewing Data

Diesel Control Unit

The display of the DCU is divided into three functional areas:

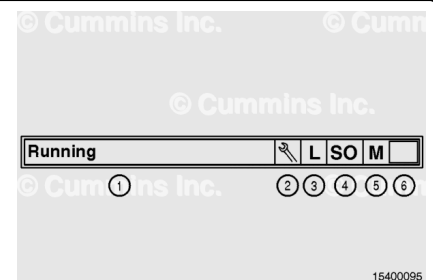
- Main screen
- Status Bar
- Soft Key Indication.

These areas are shown in the illustration.



The present status of the system is represented in the top area of the screen. The Status Bar is separated into six fields. Text and graphics are used to provide at-a-glance evaluation of the system state in these fields.

1. DCU Present Status
2. DCU Service Mode
3. DCU Local/Remote Mode
4. DCU Override Shutdown Mode
5. DCU Manual/Automatic Mode
6. DCU All OK Mode



The following tables define the possible field contents.

Field 1 of the status bar is used to display the present operating status of the DCU. A text message will be displayed that reflects the mode of operation that the DCU is performing. A list of these messages and what they represent can be found in DCU Status Messages table.

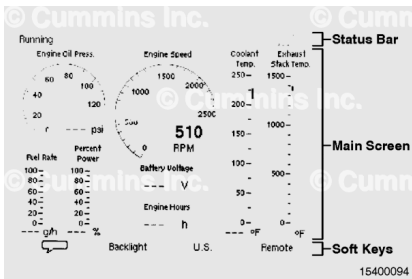
Field 1 DCU Status Messages	
Message	Comments
Ready	Ready for engine start
About to start. . .	When automatic start in Automatic mode
Prelube	Manual or Automatic Prelube
Cranking	"Cranking Attempt n" if n>1 for Automatic mode
Awaiting Run	Start request made but awaiting release
Awaiting Next Start Attempt	Delay between start attempts
Running	Engine running
Delayed Stop	Automatic stop signal received in automatic mode
Cooling	Cool down mode after delayed stop
Running?	Lost the only source of Run indication
Stopping	Stop request made
Stopped	RPM < 5
Blocked	Blocked from running

Field 1 DCU Status Messages	
Message	Comments
Extended Stopping	Pre-programmed stop delay, 50 seconds. Engine can not be started in this state
Start Disabled	

Fields 2 through 6 of the status bar are used to denote the DCU active modes. A graphical representation of the active mode(s) will be displayed in the indicator boxes. The function of each field is described below in the Status Bar Indicators table.

Field 2 - 6 DCU Status Messages		
Field	Name	Description
2	Service	Wrench - Service interval/maintenance alarm active
3	Local/Remote	L - Local start only mode active
4	Shutdown Override	SO - Shutdown Override Active
5	Manual/Automatic	M - Manual Start/Stop Mode Active
6	All OK	Green - No active or unacknowledged red/amber faults present

The most comprehensive area of the display is in the Main Screen. It is responsible for representing available engine data and all menu content. The information in this area changes with the selected function. Some information can continue off of the Main Screen area. In this case, the arrow keys on the side of the DCU can be used to advance the screen/cursor up or down. Additionally, while in the engine data viewing screens, there are multiple data templates that can be scrolled through with the use of the arrow keys.



Page Navigation

Diesel Control Unit

This button provides access into the Main Menu of the DCU. From this menu the user can access additional information from the DCU that is **not** contained in the default display screens. The user can also access troubleshooting screens and programming screens for the DCU as well. The menu is separated into several levels. From each level, a specific list of sub-levels is available. The user navigates this list of levels with a combination of Hard and Soft Key presses. A list of the available options from this menu, along with brief descriptions of their function, can be found in DCU Main Menu Option Levels table.

DCU Main Menu Option Levels				
Top Level	2nd Level	3rd Level	4th Level	Description
Screen Backlight	Automatic, 10 percent, 50 percent, 100 percent.			In Automatic, DCU uses a photo sensitive feedback from the front panel.
Language				Select Language
Units				Select Units (Metric/U.S.) Display Unit

DCU Main Menu Option Levels				
Top Level	2nd Level	3rd Level	4th Level	Description
Counters	Dialog that display counters. Option to reset trip values.			From ECM if available, from DCU if not . Shall include: - Total hours: minutes - Trip hours: minutes (since start) - Trip hour: minutes (resettable) - Start counter - Fuel consumption counters. Option to reset the fuel consumption trip counters, originated from J1939 or calculated values.
Automatic/Manual				Displays an "M" in the DCU Status Bar.
Local/Remote				Displays an "L" in the DCU Status Bar.

DCU Main Menu Option Levels				
Top Level	2nd Level	3rd Level	4th Level	Description
Prelube Override	Yes, No			Record all changes as events in the Log. If selecting "Yes" a dialog shall appear "Are you sure you want to select Prelube Override? Yes, Cancel" When in Prelube Override, continues in state CRANKING after Prelube Activation signal. When active, shall display "Prelube Override" in the LCD. When engine has been running for >10 sec, or reached Start Failure Alarm, this shall reset back to "No" (do not want to stay in Prelube Override). The configuration shall determine if this menu item is available or not.

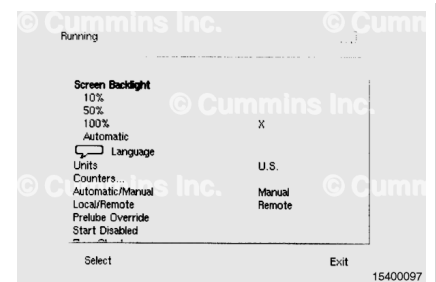
DCU Main Menu Option Levels				
Top Level	2nd Level	3rd Level	4th Level	Description
				This menu option disappears if Configuration - Prelube Installed - No.
Start Disabled				Start can be disabled in the menu or from external input. While in this mode; if user press the Start button, a dialog shall inform that: "Start Disabled". Dialog disappears after 2 sec.
Troubleshooting	Tabular list of status for all available DCU functions			Use Tab soft button to navigate status pages
Event Log	Displays the Event Log			Use soft keys to filter/sort events
Engine Service Interval	Tabular list of all programmed service events			Displays time left until programmed interval. Interval value can be programmed from Configuration option.
Connect a PC	Displays IP address of DCU. Toggle Enabled tick box and Save to activate.			Enables DHCP server in DCU to allow for communication with Service PC.

DCU Main Menu Option Levels				
Top Level	2nd Level	3rd Level	4th Level	Description
Version Information				DCU 410 Firmware, Hardware, Colibri, etc.; RIO 410 Firmware, Hardware (x4); SDU 410 Firmware, Hardware.
Configuration	Password dialog			If Password is set. In the event of a lost/forgotten password, an encrypted password is shown here that can be decrypted by Cummins Inc. or Auto-Maskin.
	Password Configuration			
	Load Configuration	Configuration Wizard		Load configuration based on wizard profile
	Interface Design	Form List	Template Design Pages	Change/Add Screen Template content
	Automatic Start/Stop Config.	Dialog with settings		Number of start attempts, etc.
	Language	English		Select the language to be used.

Screen Backlight:

The operator in manual mode can choose the backlight illumination from 10, 50, or 100 percent settings.

In Automatic mode, the DCU uses a photo sensitive feedback from the front panel.



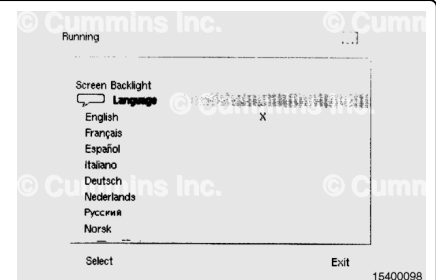
Language:

The operator can view what language the DCU will display.

Choices of language:

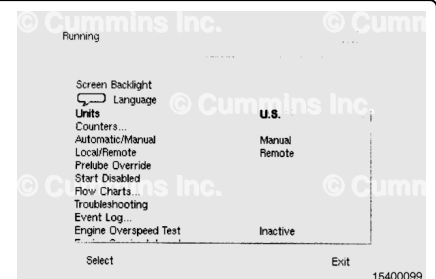
The language can be set in the DCU Power On Wizard.

Reference the setup portion of this procedure.



Units:

The operator can toggle between Metric and U.S. values in measurements.



Counters:

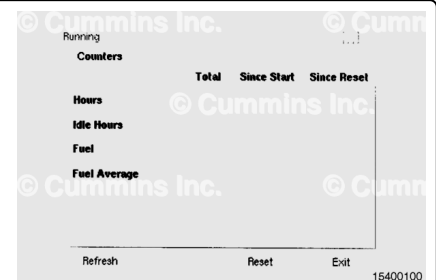
The operator has the option to reset trip values. The screen displays a dialog that displays counters.

Values are from the ECM if available and from the DCU if **not**.

The counters include:

- Total hours: minutes
- Trip hours: minutes (since start)
- Trip hour: minutes (resettable)
- Start counter
- Fuel consumption counters.

Option to reset the fuel consumption trip counters, originated from J1939 or calculated values.



Automatic/Manual:

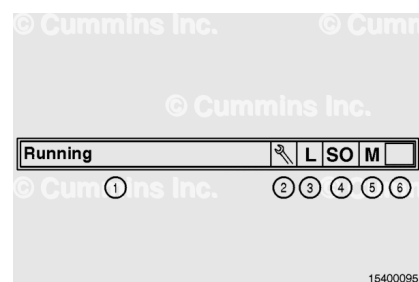
The Automatic/Manual setting is displayed in field (5) of the status bar.

The Select button will toggle between Manual and Automatic.

Local/Remote:

The Local/Remote setting is displayed in field (3) of the status bar.

The Select button will toggle between Local and Remote.



Prelube Override:

The operator can toggle between Yes and No.

Record all changes as events occur in the Log.

If selecting "Yes" a dialog shall appear "Are you sure you want to select Prelube Override? Yes/Cancel"

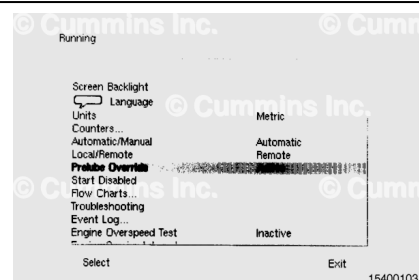
When in Prelube Override, continues in state CRANKING after Prelube Activation signal.

When active, shall display "Prelube Override" in the LCD.

When engine has been running for >10 sec, or reached Start Failure Alarm, this shall reset back to "No" (do **not** want to stay in Prelube Override).

Note : The configuration shall determine if this menu item is available or not, if configured with no prelube installed. the menu will **not** appear.

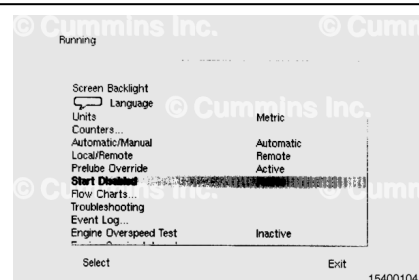
This menu option disappears if Configuration - Prelube Installed - No.



Start Disabled:

Start can be disabled in the menu or from an external input.

While in this mode, if the user presses the Start button, a dialog shall inform that: "Start Disabled". The dialog disappears after 2 seconds.



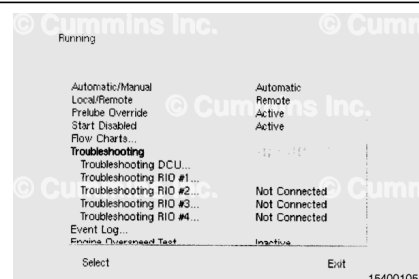
Troubleshooting:

Displays a tabular list of status for all available DCU functions.

Use the tab button to navigate through the screen displays.

The functions are:

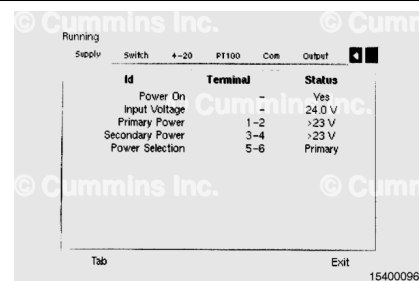
- Troubleshooting DCU
- Troubleshooting RIO 1



- Troubleshooting RIO 2
- Troubleshooting RIO 3
- Troubleshooting RIO 4.

The tab screens for the troubleshooting functions are:

- Supply
- Switch
- 4-20 mA
- PT100
- Com
- Output
- Input
- Run Sources
- Button Test.



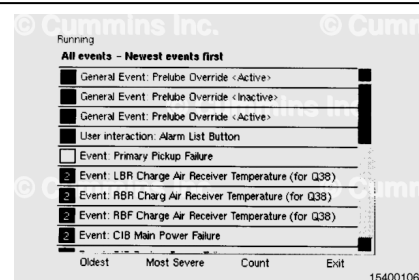
The tab screens are the same for all functions.

Event Log:

The operator can use the soft keys to filter or sort events in this mode.

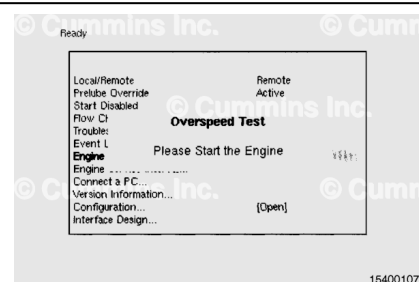
The tab screens are:

- Oldest
- Newest
- Most Severe.



Engine Overspeed Test:

The operator can configure the class vessel to allow for this test.



Engine Service Interval:

This is a tabular list of all programmed service events. It displays time left until programmed interval. The interval value can be programmed for the Configuration Menu in this procedure.

The tab screens are:

- Service 1
- Service 2



- Service 3
- Service 4.

The operator can confirm the maintenance interval by pressing the Service Done button.

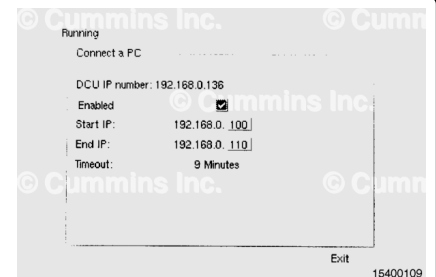
If the Service Done button is pressed, the maintenance interval will advance to the next interval.

Connect to PC:

Displays the IP address of the DCU.

Toggle Enabled tick box and Save to activate.

Enables DHCP server in DCU to allow for communication with Service PC.



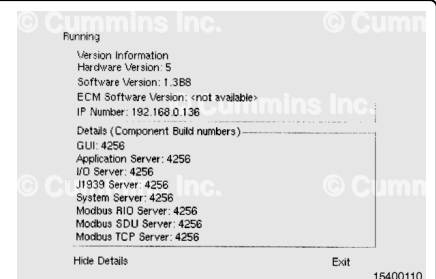
Version Information:

Displays the following information for the operator:

- Hardware Version
- Software Version
- IP Number.

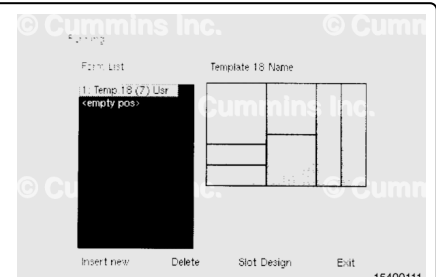
Details can be displayed at anytime by pressing the Detail button. The details include:

- DCU 410 Firmware, Hardware, Colibri, etc.
- RIO 410 Firmware, Hardware (x4)
- SDU 410 Firmware, Hardware.



The screen illustrates template design pages and forms list.

The operator can change user defined screens and other screens can be adjusted.



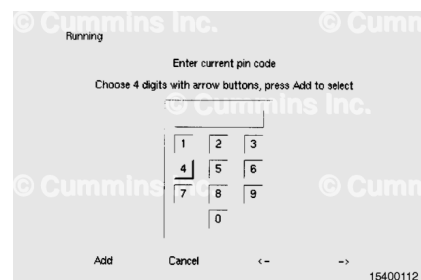
Configuration Menu

The following steps are for DCU configuration.

This screen displays dialog for the password.

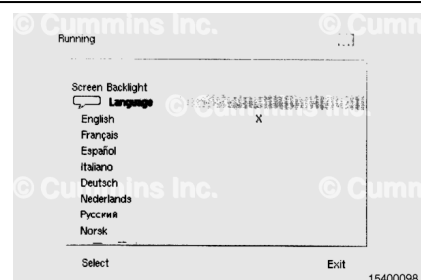
In the event of a lost or forgotten password, an encrypted password can be obtained from a Cummins® Authorized Repair Location.

This directs the operator to the Configuration Menu.



The operator can set which language the DCU will display by using the up and down arrow keys.

The default language is English.

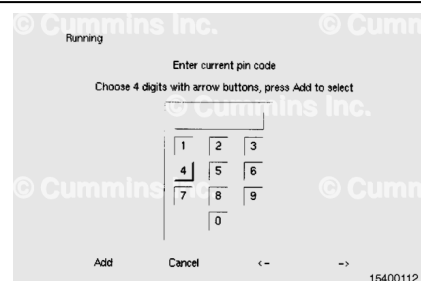


The following steps are Miscellaneous under the Configuration Menu.

The operator **must** enter a pass code before the Miscellaneous configuration parameters can be changed.

The parameters include:

- Automatic Buzzer Off
- Fuel Consumption
- Speed Relays
- Counter Source
- Start/Stop Buttons
- Network Configuration
- Factory Reset.

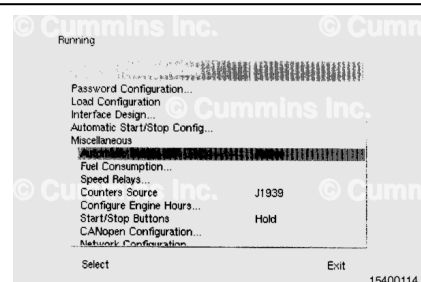


Automatic Buzzer Off:

The automatic buzzer times out after 12 hours for commissioning.

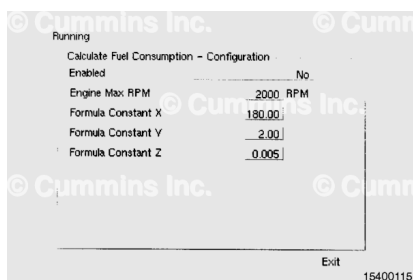
The internal (and external if configured) buzzer will sound once **only** and then turn off.

The operator can toggle between inactive and active.



Fuel Consumption:

Cummins Inc. recommends adjusting Calculate Fuel Consumption parameters. Fuel consumption inaccuracies can occur.

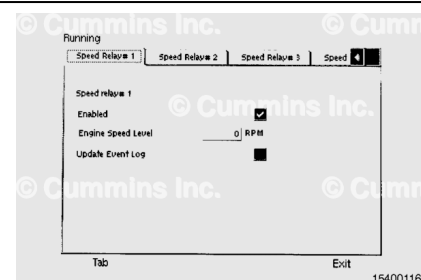


Speed Relays:

The operator can enable one or all four of the speed relay outputs.

- Enabled
- Engine speed level
- Update Event Log.

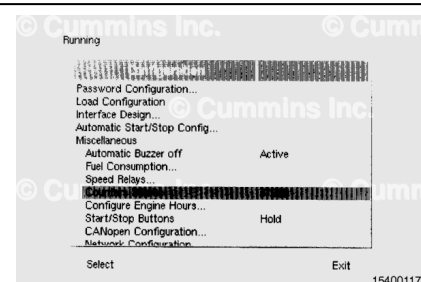
The settings are the same for all four relays.



Counter Source:

Cummins Inc. recommends this parameter to be set to J1939. The DCU will be able to read input from the ECM.

Engine hours configuration lets you offset broadcast J1939 hours locally on the display.



Start/Stop Buttons:

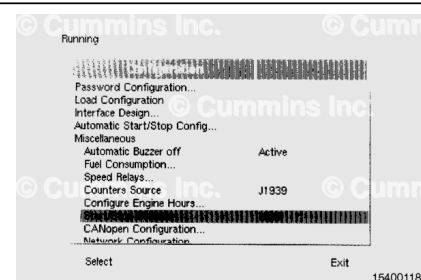
The operator can toggle between Latched and Hold modes by pressing the Select button.

Hold to Start:

To start the engine, the operator must press and hold the start button until the engine is running. When the engine speed has reached 500 rpm, the engine run signal will become active and the starter will automatically be disengaged. If the engine speed falls below 100 rpm, the run signal will become inactive, thus allowing another start attempt. If the start button is released at any point prior to the engine running, the start process will terminate.

Latched Start:

To start the engine, the operator first must press the start button. Upon pressing the start button, the DCU will prompt the



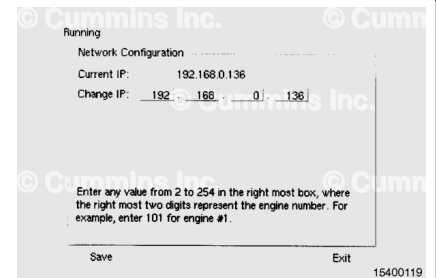
operator with a start confirmation message “Are you sure?” The operator must then press the soft button labeled “Start” to begin the start sequence. The operator is also offered a “Cancel” soft button in the event the start button press was accidental.

Network Configuration:

Enter any value from 2 to 254 in the right-most tick box, where the right-most two digits represent the engine number.

For example: Enter 101 for engine number 1.

- Current IP - save changes
- Change IP - use Save button.

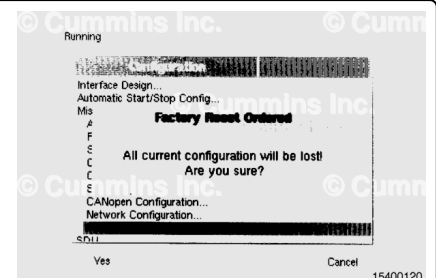


⚠ CAUTION ⚠

All present configuration settings will be lost by performing a Factory Reset.

Factory Reset:

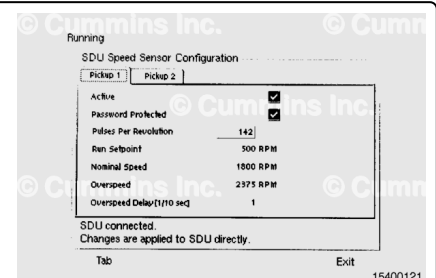
Removes the present configuration and restores the factory default settings. The DCU will startup the first power sequence at the next power-up.



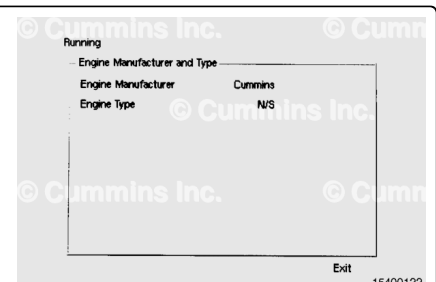
The operator can configure the DCU source(s) for the speed sensor rpm by using the Select button.

- Speed Sensor Configuration
- Pickup 1 pulse/rev
- Pickup 2 pulse/rev
- Update DCU -> SDU
- Update SDU -> DCU.

A note will be displayed stating changes are applied to the SDU directly.



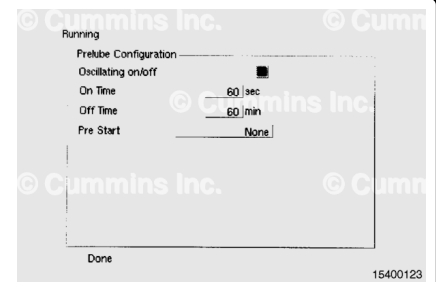
The screen displays present values of the Engine Manufacturer/Type.



The operator can enable or set the prelube parameter.

The following parameters can be set:

- Oscillating ON/OFF - check tick box
- ON time
- OFF time
- Pre Start
- Until Pressure Timeout
- Action Upon Timeout
- Override Allowed - check tick box.

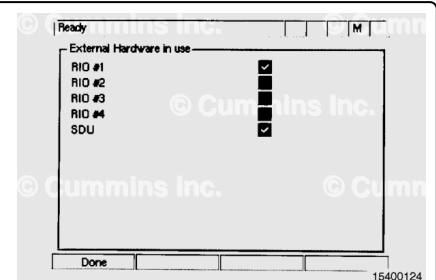


The operator can choose how many minutes between prelube cycles when checking the tick box on the oscillating prelube. This function can also speed up the prelube at the start of the engine.

The operator can enable external components (i.e. RIO, SDU, etc.) through the external hardware option screen display.

External hardware in use:

- RIO1 - if active, a tick box will appear
- RIO2 - if active, a tick box will appear
- RIO3 - if active, a tick box will appear
- RIO4 - if active, a tick box will appear
- SDU - if active, a tick box will appear.



The C Command Elite Plus™ system CIB will have a single RIO and SDU installed.

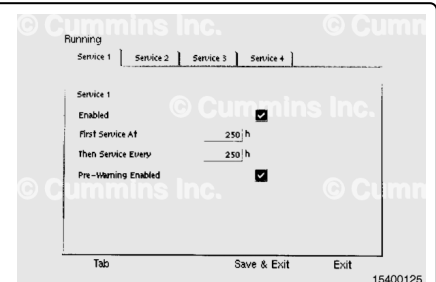
The operator can enable/setup Service Interval Messages by selecting one of four service interval tabs:

Service 1

The following parameters are the same on all four service intervals.

- Enabled - check tick box
- First Service At
- Then Service Every
- Pre-Warning Enabled - check tick box
- SDU - if active, a tick box will appear.

Service 2, 3, and 4 can be setup like Service 1.



The operator can enable/setup I/O channel for the DCU display and/or monitoring.

The I/O options menu contains the following parameters:

Engine Speed - D + R (DCU and RP)

- Sensor
- Display
- Alarming
- Other
- Preview.

Switch Inputs

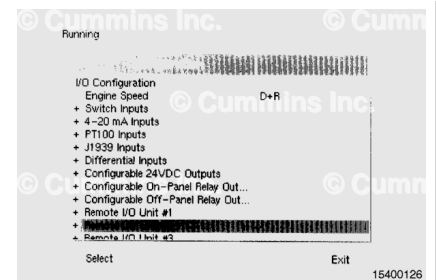
- Switch Inputs 1 through 8 (E = Event Log is active)

4-20 mA Inputs (tab screens are the same as Engine Speed)

PT100 Inputs (PT100 1 through 4) (tab screens are the same as Engine Speed)

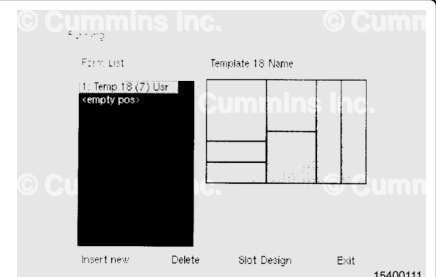
J1939 Inputs

- Percentage Load - D + R (DCU and RP)
- Air Filter Differential
- Air Inlet Pressure
- Auxiliary Pressure
- Barometric Pressure
- Battery Voltage
- Boost Pressure.



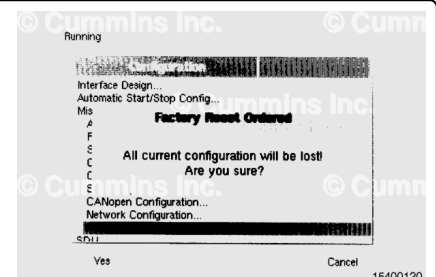
The operator can use the Template Design Pages and change/add screen template content. The operator can choose the default template or create an operator defined template.

Note : The display will view operator defined screens only.



Factory Reset

The operator can reset the parameters values to the factory default settings.



Connect to PC

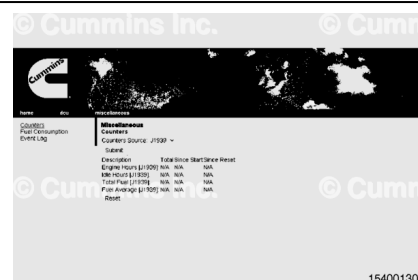
The following steps are for DCU main menu connecting to a PC with website interface.

The website interface for the DCU will configure the remote panel parameters when the files are uploaded. The remote panel uses the DCU parameters set by the operator on the Connect to PC menu.



Counters

A list of data the DCU receives from the J1939 data link.



Prelube Override

This function is set by default by the manufacturer.



Start Disabled



Troubleshooting

The following parameters can be set:

- Supply
- Switch
- 4-20 mA
- PT100
- Communication
- Output
- Input
- Run Sources.

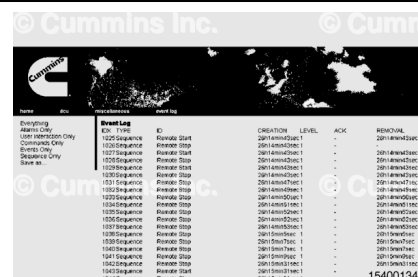


For example: The supply values are shown in the illustration.

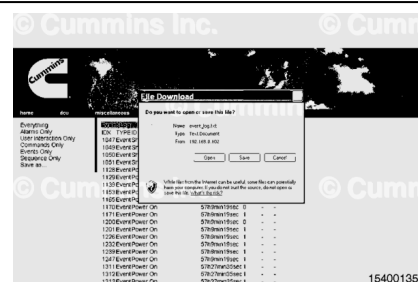
Event Log

There are several parameters the operator can view from this function:

- All Events
- Alarms **Only**
- User Interaction **Only**
- Commands **Only**
- Sequence **Only**.



The operator can save this file by choosing Save on the menu.



Engine Service Interval



Version Information



Configuration

The operator can upload the file once configured.



The screenshot shows the Cummins iQMT software interface. At the top, there is a navigation bar with 'Home' and 'File' buttons. Below the navigation bar, there is a large image of a Cummins engine. In the foreground, a dialog box titled 'Load Factory Default Configuration File' is open. The dialog box contains a list of available configuration files, each with a file name and a description. The list includes:

- 015 - Net Heat Exchanger - Type Approved - Aux 60Hz
- 015 - No Heat Exchanger - Type Approved - Aux 60Hz
- 015 - No Heat Exchanger - Net Type Approved - Propulsion
- 008 - No Heat Exchanger - Net Type Approved - Aux 50Hz
- 028 - No Heat Exchanger - Net Type Approved - Aux 60Hz
- 018 - Heat Exchanger - Type Approved - Propulsion
- 008 - No Heat Exchanger - Type Approved - Aux 60Hz
- 020 - No Heat Exchanger - Type Approved - Aux 60Hz
- 015 - Heat Exchanger - Net Type Approved - Aux 60Hz
- 015 - Heat Exchanger - Net Type Approved - Aux 60Hz
- 015 - No Heat Exchanger - Net Type Approved - Propulsion
- 015 - No Heat Exchanger - Net Type Approved - Propulsion
- 015 - No Heat Exchanger - Net Type Approved - Propulsion

The dialog box also has a 'Factory Default User Updated' button and a 'Load' button.

[illegible]

The language setting defaults to English.

The operator can change the language for all channels or specific channels.



The following parameters can be set using the Miscellaneous menu:

- Auto Buzzer Off - The operator can toggle between Yes and No.



- Fuel Consumption



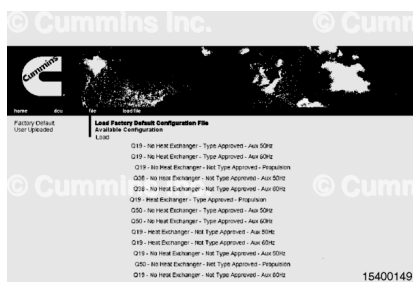
- Counters Source



- Start/Stop Buttons- The operator can set the button to hold for start and stop.



- Factory Reset



The operator can set the parameters on the speed sensor inputs to the SDU.



The operator can change the engines name and view which type of engine application and manufacturer. The manufacturer **always** defaults to Cummins Inc.

The DCU name is also the engines name.



The operator can set the parameters of the prelube start and stop.



The operator can set the parameters for the speed sensors to the DCU.



The screen displays which type of hardware and which software version is being used.



The operator can change service intervals by choosing the submit soft button when parameters have been set.



The I/O configuration consists of the following parameters:

- Engine Speed Inputs
- Switch Inputs
- 4-20 mA Inputs
- PT100 Inputs
- J1939 Inputs
- Differential Inputs
- 24-VDC Input and Outputs
- Special Input (used by OEM)
- On-Panel Relay Outputs
- Off-Panel Relay Outputs
- Remote I/O Unit (Speed Relays).



Only a few of the parameters are shown in the following illustrations. All parameter choices remain the same.

- 4-20 mA Inputs



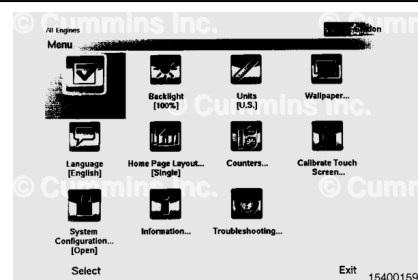
- 24-VDC Inputs or Outputs - The operator can set up to four 24-VDC outputs.



Remote Panel

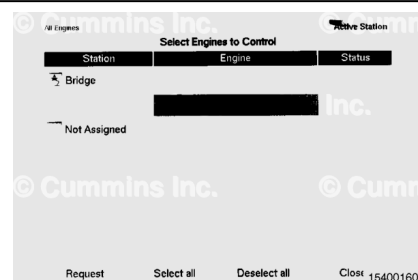
The Remote Panel (RP) main menu consist of the following functions:

- Active Station
- Backlight
- Units
- Wallpaper
- Language
- Calibrate Touch Screen
- System Configuration
- Information
- Troubleshooting.



Active Station:

The operator can view which station is active or non active by looking in the upper right hand corner of the RP display. The operator can request this station to be Active.



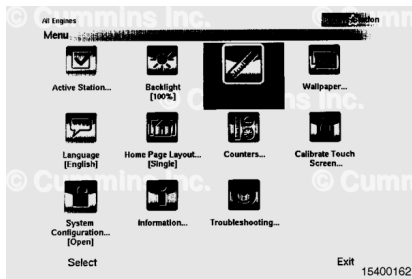
Backlight:

The operator can adjust the intensity of the backlight. This process is the same as the DCU.



Units:

The operator can toggle between Metric and U.S. measurement values.



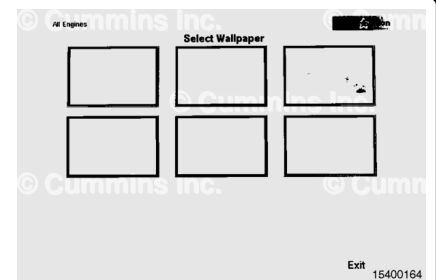
Language:

The language setting defaults to English.



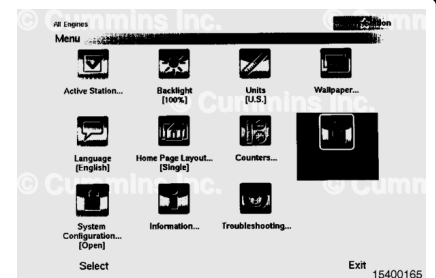
Wallpaper:

The operator can choose between 1 of 3 wallpapers for the display.



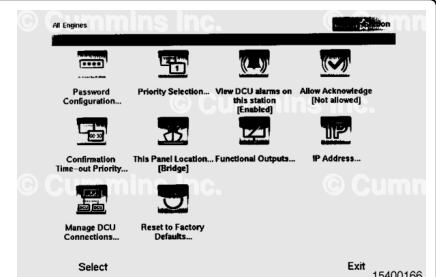
Calibration Touch Screen:

The operator can calibrate the touch screen by touching the tick boxes on the screen. The tick boxes are located in each corner and the center of the display.



System Configuration:

For the operator to change the system configuration, a password **must** be entered (default 1234). If the password is lost or forgotten, contact a Cummins® Authorized Repair Location to regain access. The following parameters can be adjusted:

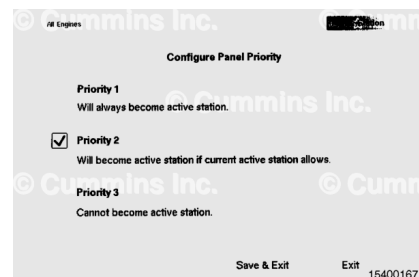


- Password
- Priority Selection
- Alarms
- Acknowledgement of Alarms
- Confirmation Timeout Behavior

- Panel Location
- Date and Time
- IP Address
- DCU Connections
- Factory Reset.

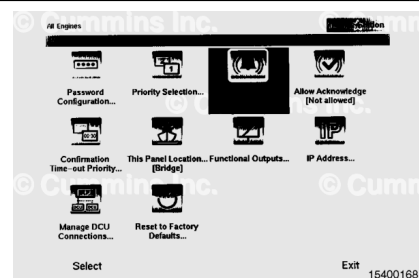
Priority Selection

The operator can give a brief description of each 1, 2, or 3. Active station priority capabilities are described on the selection screen.



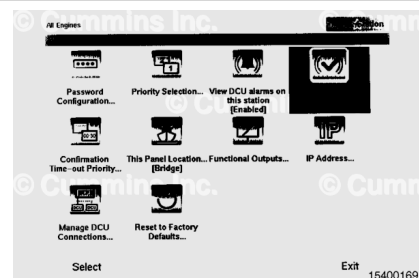
Alarms

The operator can disable all the alarms at this active station Yes [No]. Selecting Yes inhibits all alarms on this station.



Acknowledgement of Alarms

Alarms can not be silenced or acknowledged on the remote panel, but must be silenced and acknowledged on the DCU at the engine.



Confirmation Timeout Behavior

The operator can set the timeout to 10 to 255 seconds [10]. The default is hand over control upon timeout. The operator can deny transfer upon timeout.

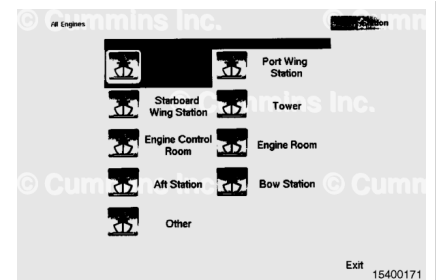


Panel Location

This station can be named the following:

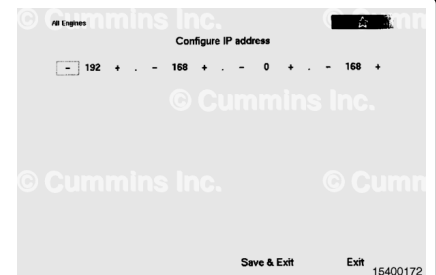
- Bridge

- Port Wing Station
- Starboard Wing Station
- Tower
- Engine Control Room
- Engine Room
- Aft Station
- Bow Station
- Other



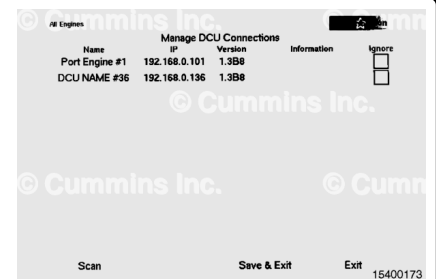
IP Address

The operator can change the IP address of the RP in the network.



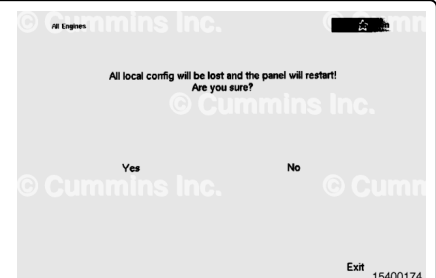
DCU Connections

This screen is dialog that displays scan process and results. List available DCU CIB panels in the network, and select (check the tick box) for each. The default setting for this parameter is "none are ignored". Specific DCUs in the network can be ignored.



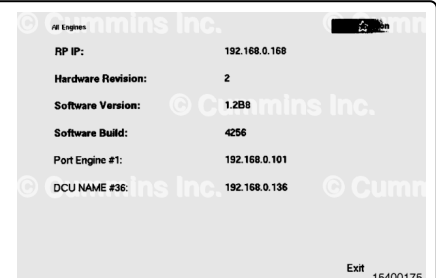
Factory Reset

The operator can reset the configuration to default manufacturer settings.



Version Information:

The Version Information can be displayed by the operator.



Troubleshooting:

The operator can use the Troubleshooting from the main menu to view different parameters of the engine.

The tab screens consist are:

- **Supply**
- **Switch**
- **4-20 mA**
- **PT100**
- **Communication**
- **Output**
- **Input**
- **Run Sources**
- **Button Test.**

The screenshot shows the 'Troubleshooting' menu with 'Communication' selected. It displays a table with columns: Description, Terminal, Status, M, Terminal Name, Tx, Rx, and Status. The 'Supply' tab is active, showing input voltage at 23.8 V. Below this, there are sections for 'Inputs' and 'Outputs' with their respective terminal numbers and statuses.

Supply:

This screen displays the power source to the RP.

The screenshot shows the 'Supply' tab selected. It displays a table with columns: Id, Terminal, and Status. The table lists power-related parameters such as Power On, Input Voltage, Primary Power, Secondary Power, and Power Selection, along with their terminal numbers and current status.

Switch:

This screen displays the eight switch inputs to the SDU unit. These switches are normally open.

The screenshot shows the 'Switch' tab selected. It displays a table with columns: Id, Terminal, and Status. The table lists eight switch inputs (Switch Input #1 through #8) with their terminal numbers and current status (Open or Closed).

4-20 mA:

The 4-20 mA circuits are typically for pressure sensors. This screen displays each one of the 4-20 mA circuits and whether they are in use or **not** in use.

The screenshot shows the '4-20' tab selected. It displays a table with columns: Id, Terminal, and Status. The table lists four 4-20 mA circuits (4-20mA #1 through #4) with their terminal numbers and current status (e.g., 930.05 psi, Not in use).

PT100:

The PT100 circuits are typically for temperature sensors. This screen displays the status of each circuit.

Ready Cummins Inc. L Cummins Inc.		
Supply	Switch	4-20 PT100 Com Output
Id Terminal Status		
PT100 #1	32-34	180 F
PT100 #2	35-37	693 F
PT100 #3	38-40	Not in use
PT100 #4	41-43	Not in use
Tab Exit 15400180		

Com:

This screen displays the communication link between the SDU and multiple RIO circuits to the RP display.

Ready Cummins Inc. L Cummins Inc.					
Supply	Switch	4-20	PT100	Com	Output
Id Terminal Name Tx Rx Status					
COM 1	60-62	SDU	280422	420634	On Line
COM 2	57-59	RIO S#1	218741	724589	On Line
COM 2	57-59	RIO S#2			Not in use
COM 2	57-59	RIO S#3			Not in use
COM 2	57-59	RIO S#4			Not in use
COM 3	52-56	Modbus RTU			Off Line
COM 4	49-51	J1939	385124	478298	On Line
COM 6	RJ45	Modbus/TCP			Off Line
Tab Reset Exit 15400181					

Output:

This screen displays the Inactive or Active status of each output circuit.

Ready Cummins Inc. L Cummins Inc.		
Supply	Switch	4-20 PT100 Com Output
Id Terminal Status		
On Board Relay #1	63-65	Inactive
On Board Relay #2	66-68	Inactive
User Config Output #1	69	Active
User Config Output #2	70	Inactive
User Config Output #3	71	Inactive
User Config Output #4	72	Inactive
Alarm Relay	73-75	Active
Energize to Shutdown	79	Inactive
Energize to Stop	80	Inactive
Energize to Run	81	Inactive
Engine Running	82	Inactive
Crank	83	Inactive
Tab Exit 15400182		

Input:

This screen displays the Inactive or Active status of each input circuit.

Ready Cummins Inc. L Cummins Inc.		
Supply	Switch	4-20 PT100 Com Output Input
Id Terminal Status		
Prelube Complete Switch	86	Inactive
Start Disabled Switch	87	Inactive
Automatic/Manual Switch	88	Inactive
PMS Start Switch	89	Inactive
PMS Stop Switch	90	Inactive
Remote Start Switch	91	Inactive
Remote Stop Switch	92	Inactive
Remote Acknowledge Switch	93	Inactive
Shutdown Override Switch	94	Inactive
In Gear Switch	95	Inactive
User Config Switch Input #1	96	Inactive
User Config Switch Input #2	97	Inactive
Tab Exit 15400183		

Run Sources:

This screen displays the status on each source circuit and whether the function has been configured or **not**.

Ready Cummins Inc. L Cummins Inc.		
4-20	PT100	Com Output Input Run Sources
Source Configured Status		
Engine Speed (DCU)		
Engine Speed (SDU)	X	
Engine Speed (J1939)	X	
Switch Input #1		
Switch Input #2		
Switch Input #3		
Switch Input #4		
Switch Input #5		
Switch Input #6		
Switch Input #7		
Switch Input #8		
Tab Exit 15400184		

Button Test:

This screen displays the status of each one of the buttons.

Ready					L				
00	Com	Output	Input	Run Sources	Button Test				
Button					Status				
Alarms Button					Released				
Up Button					Released				
Menu Button					Released				
Down Button					Released				
Func 1 Button					Released				
Func 2 Button					Released				
Func 3 Button					Released				
Func 4 Button					Released				
Stop Button					Released				
Start Button					Released				
Hold down Func 2 to reveal Tab and Exit button									
Func 1		Func 2		Func 3		Func 15400185			